Letlhogonolo Fanampe

AI Solutions Engineer

Pretoria, South Africa | Remote and hybrid | Available immediately

l.fanampe@gmail.com | +27 83 483 7699

djpapzin.com | LinkedIn | GitHub

Profile

AI Solutions Engineer building LLM applications that connect business knowledge, data and workflows. Hands-on experience delivering RAG and SQL assistants, API integrations, OCR pipelines and Linux deployments. Combines application engineering with AI response evaluation and human feedback for model training.

Core skills

Python, SQL, FastAPI, LangChain, Flask, Django, Streamlit, REST APIs, OpenAI APIs, Together AI, RAG, NLP, LLM fine-tuning, FAISS, DeepLake, ChromaDB, SQLite, Docker, Git, Linux, Nginx, systemd, OCR, RLHF and model evaluation.

Professional experience

Freelance AI Engineer

Self-employed | Remote | Feb 2024 - Present

- Build AI applications and developer tools using Python, language models and external APIs, from early prototypes through integration and demonstrations.
- Developed VocalThread for YouTube comment analysis and speech generation, and built local recovery and agent workflow tools described in the selected projects below.

AI and Machine Learning Engineer

Kwantu Pty Ltd | Contract | Remote | Oct 2024 - Apr 2025

- Developed and deployed RAG assistants for PDF documents and SQL databases with Python, LangChain, OpenAI embeddings, FAISS and DeepLake, including multilingual retrieval.
- Integrated chatbot services with FastAPI, WhatsApp Business API and Telegram using models and APIs including Mixtral and Together AI.
- Built an ID-document processing pipeline covering data cleaning, Label Studio annotation, Detectron2 training and OCR with Tesseract and PaddleOCR.
- Maintained CentOS/Linux deployments with systemd and Nginx; resolved SQL schema mismatches, API rate limits, connectivity problems and inconsistent model responses.

AI Training Data Specialist

Outlier via G2i | Contract | Remote | Jul 2024 - Aug 2025

- Evaluated and ranked model responses for accuracy, coherence, relevance and adherence to guidelines; verified training data and reviewed peer submissions.
- Designed prompts and adversarial tests to identify reasoning errors and model failure cases, and documented feedback for reinforcement learning from human feedback.

Professional experience continued

AI Prompt Evaluator

Translated | Contract | Remote | Oct 2024 - Dec 2024

- Fact-checked, rated and rewrote AI-generated prompts and responses for a major GPU enterprise client, documenting accuracy, relevance and instruction-following issues.

Selected AI projects

Agent Handoff Kit

Built a Python and SQLite prototype that resumes interrupted local work from saved checkpoints. Uses transactional receipts, leases and generation fencing to suppress duplicate effects within a local database transaction. Published a recovery demo, architecture notes, source code and tests.

Project and source

AgentOps Mobile Command Center

Built a FastAPI and SQLite control room for goals, approvals, run history and model routing through Telegram. Includes Docker setup, a demonstration mode and optional integrations with model providers.

Project and source

TruthGuard

Contributed as AI Specialist to a team news-classification project using Llama 3 and Together AI. The project includes fine-tuning on the WELFake dataset and a Python and Flask classification API.

Project and source

VocalThread

Developed a Streamlit application that analyses YouTube comments and turns them into spoken dialogue using ElevenLabs, YouTube Data API and Google Gemini. Features include sentiment analysis and chat with comments.

Project and source

Earlier experience

Laboratory Analyst and Assistant

AfriSam | Lichtenburg, South Africa | Dec 2014 - Jul 2022

Performed routine and specialised laboratory testing, maintained analytical equipment and supported product quality assurance. Built a foundation in data analysis, troubleshooting and consistent evaluation procedures.

Education and professional development

National Diploma in Analytical Chemistry | Tshwane University of Technology | 2010 - 2015

LangChain and Vector Databases in Production | Activeloop | Sep 2023

Master Python 3 | Udemy | Aug 2023

Career Essentials in System Administration | Microsoft and LinkedIn | Mar 2023

Recognition

WebGPU Hackathon One-Day Challenge winner with VisualPro | Aug 2023. Built Node.js API endpoints and SQLite storage for a team 3D data visualisation application.